

WENJING TIAN

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EDUCATION

Nanyang Technological University (NTU) | Master of Science in Information Systems | Expected 2027

University of California, Los Angeles (UCLA) | B.S. in Cognitive Science, Computing Specialization | Minor in Digital Humanities | 2025

EXPERIENCE

PetroChina Planning & Engineering Institute, Energy Conservation Center | AI Agent Development & Data Governance Intern | Beijing, China | Sep 2025 – Jun 2026

Enterprise AI Review Platform & Multi-Agent Architecture

- Designed and implemented a three-stage Multi-Agent review workflow spanning information extraction, logic validation, and review-comment generation, decomposing complex business review processes into modular agent tasks with human-in-the-loop checkpoints.
- Built a domain-specific knowledge base using RAG, integrating regulations, industry standards, and historical reports; combined LLM-based generation with rule-based validation to improve factual and logical consistency.
- Developed automated content extraction, policy compliance validation, and energy-efficiency/carbon-emission benchmarking modules with dynamic technical catalog and knowledge-base retrieval, contributing to a 60%+ improvement in internal review efficiency.

Carbon Reduction Data Governance & BI Platform

- Governed multi-source carbon reduction project data across CCER, VCS, and Gold Standard (GS) mechanisms, performing data cleaning, field standardization, quality validation, and structured data modeling.
- Developed an NLP-based location extraction pipeline using open-source administrative-division dictionaries to identify and populate province/city information from unstructured project descriptions, reducing manual data-processing effort.
- Independently designed a web-based data product prototype and built an interactive SugarBI dashboard for querying and analyzing project, policy, methodology, and carbon-market transaction data.

Astral Culture Media

 | Marketing & Web Design Intern | Seattle, WA | Oct 2024 – Jan 2025

- Analyzed the website's lead-generation funnel and user behavior to identify key drop-off points; redesigned interaction flows and contact forms, increasing lead conversion by 15%.
- Researched 50+ competitors across traffic acquisition, content strategy, and user interaction patterns; supported landing-page redesign and SEO keyword optimization, with A/B testing contributing to a 25% increase in organic search traffic.

KZE Group

 | Data Analyst Intern | Houston, TX | May 2024 – Sep 2024

- Cleaned and analyzed data from 500+ TikTok creator accounts, identifying eight key engagement behavior patterns to support creator profiling, selection, and advertising strategy.
- Conducted quantitative analysis of user behavior, engagement rates, and campaign performance to optimize creator matching, contributing to a 25% increase in ROI and a 20% reduction in cost per engagement.

PROJECTS

Shopping Copilot — Conversational Search & Recommendation AI Agent

 | TikTok TechJam Hackathon 2026 | Python, LLMs, Embeddings, Information Retrieval

- Developed a multi-turn conversational shopping agent for a 50K-product catalog, converting natural-language requests into structured intents, preferences, and product constraints while dynamically selecting clarification, retrieval, reranking, exploration, and recommendation strategies.
- Designed dialogue-state management for Buying, Browsing, Intent Override, and Boundary scenarios, continuously tracking hard constraints, preferences, exclusions, and intent changes to prevent outdated context from degrading recommendations.
- Built a BM25 + embedding hybrid retrieval pipeline with structured constraint filtering and evidence-based reranking, dynamically adapting retrieval strategies based on confidence, constraint coverage, and recommendation stagnation.
- Established an agent evaluation framework using HitRate@K, MRR, and MTTC, with robustness testing across paraphrases, typos, short responses, and negative constraints plus ablation studies; achieved 99.5% HitRate@10 on the public evaluation set.

Honkai: Star Rail Dialogue NLP Analysis

 | Python, NLP, LDA, Machine Learning

- Scraped and cleaned 50K+ lines of game dialogue, performing text preprocessing, tokenization, stop-word filtering, and feature engineering to construct a structured narrative corpus.
- Applied LDA topic modeling and sentiment analysis to identify chapter-level themes, character language patterns, and narrative shifts, visualizing topic evolution across the storyline.
- Built a text classification model to automatically predict dialogue chapters from linguistic features and produced a data-driven narrative analysis report.

HCI-Based Customer Service Chatbot Design & Evaluation

 | HCI, Chatbot, A/B Testing, UX Research

- Designed two chatbot interaction strategies for an e-commerce customer-service scenario and developed interactive web prototypes.
- Defined evaluation metrics including task completion rate, cognitive load, and user satisfaction; conducted an A/B experiment to compare conversational patterns and derive chatbot UX recommendations.

SKILLS

AI & Machine Learning: LLM Applications, RAG, AI Agents, Embeddings, Information Retrieval, NLP, scikit-learn **Programming & Data:** Python, SQL, Pandas, NumPy, ETL, Excel, Git, GitHub

Analytics & BI: EDA, Statistical Analysis, KPI Analysis, Data Validation, Tableau, Power BI, SugarBI **Databases:** MongoDB, Redis